



SCOPE OF ACCREDITATION TO ISO/IEC 17025-2005
& ANSI/NCSL Z540-1-1994

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CALIBRATION

Valid To: April 30, 2017

Certificate Number: 2127.03

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following Calibrations :

I. Electrical – DC/Low Frequency

Parameter/Equipment	Range	CMC ^{2, 5} (±)	Comments
DC Voltage ³ – Measure	10 mV to 1000 V	0.0060 % of reading + 0.0035 % of range	DMM
LISN Series Voltage Drop	10 mV to 100 V	0.0084 % of reading + 0.0035 % of range	DMM
DC Voltage ³ – PQT ³ Output Voltage ESD Simulator ³ Current (1 to 30) kV, Contact Discharge	1 mV to 80 V Up to 260 V DC (0.07 to 148) A	3.6 % 3.6 % 3.4 %	Oscilloscope IEC/EN 61000-4-11 oscilloscope IEC/EN 61000-4-2, IEC/EN 61000-4-2 (1995), ISO 10605, oscilloscope using IEC ESD target

Parameter/Equipment	Range	CMC ^{2, 5} (±)	Comments
DC Voltage ³ (cont) – ESD Simulator ³			
Voltage Indication,	(1 to 30) kV	0.28 %	IEC/EN 61000-4-2, IEC/EN 61000-4-2 (1995), high voltage meter
Voltage Indication, Contact Discharge	(1 to 30) kV	2.5 %	ISO 10605, oscilloscope
EFT/Burst Generator ³ (50, 1 000) Ω Load			
Voltage	(0.1 to 5) kV	3.2 %	IEC/EN 61000-4-4, oscilloscope
Surge Generator ³			
Open and Short Circuit			
Voltage	(0.001 to 8) kV	3.3 %	IEC/EN 61000-4-5, IEC/EN 61000-4-5 (2005), IEC/EN 61000-4-12 oscilloscope
Current	(0.001 to 4) kA	3.3 %	IEC/EN 61000-4-5, IEC/EN 61000-4-5 (2005), IEC/EN 61000-4-12, oscilloscope
PQT ³			
Inrush Current	(0.001 to 4) kA	3.3 %	IEC/EN 61000-4-11, oscilloscope
Transient Immunity ³			
Surge Pulse and Load Dump Pulse Peak Amplitude Burst Pulse Peak Amplitude	(1 to 1000) V	3.5 %	ISO 7637-2, ISO 7637-2 (2004), ISO 16750-2, oscilloscope

Parameter/Range	Frequency	CMC ^{2, 5} (±)	Comments
DC Current ³ – Measure	10 mA to 3.0 A	0.14 % of reading + 0.020 % of range	DMM
Resistance ³ – Measure	0 Ω to 100 kΩ	0.017 % of reading + 0.0040 % of range	DMM 4-wire measurement
AC Voltage ³ – Measure			
100.0000 mV (Measure 0.00001 mV to 1000.000 mV)	10 Hz to 20 kHz	0.07 % of reading + 0.04 % of range	DMM
	(20 to 50) kHz	0.15 % of reading + 0.05 % of range	
	(50 to 100) kHz	0.69 % of reading + 0.08 % of range	
(1.000000 to 750.000) V	10 Hz to 20 kHz	0.07 % of reading + 0.03 % of range	DMM
	(20 to 50) kHz	0.14 % of reading + 0.05 % of range	
	(50 to 100) kHz	0.70 % of reading + 0.08 % of range	
(1 to 100) V	10 Hz to 100 kHz	3.3 %	Oscilloscope, HV differential probe
PQT ³			
Output Voltage	Up to 260 V (50 to 60) Hz	3.3 %	IEC/EN 61000-4-11, oscilloscope

Parameter/Range	Frequency	CMC ^{2, 5} (±)	Comments
AC Current ³ – Measure			
10 mA to 1.0 A	10 Hz to 5 kHz	0.12 % of reading + 0.04 % of range	DMM
(1.0 to 3.9) A	10 Hz to 5 kHz	0.17 % of reading + 0.06 % of range	
10 mA to 100 A	10 Hz to 5 kHz	3.3 %	Oscilloscope, current probe
(0.1 to 600) A	40 Hz to 1 kHz	2.4 % + 5 digit	IEC/EN 61000-4-8, clamp-on tester

II. Electrical – RF/Microwave

Parameter/Range	Frequency	CMC ^{2, 4, 5} (±)	Comments
RF Power ³ – Measure			
(-45 to 40) dBm	9 kHz to 3 GHz	0.6 dB	Power meter
Harmonics ³	9 kHz to 26.5 GHz	1.1 dBc	Spectrum analyzer
Amplitude Modulation ³ – (0 to 100 %)	50 Hz to 100 kHz	1.1 dB	Spectrum analyzer
	50 Hz to 100 kHz	3.5 %	Oscilloscope

Parameter/Range	Frequency	CMC ^{2, 4, 5} ($\pm$)	Comments
RF Insertion Loss ³ –			
Attenuators, RF Cables	5 Hz to 9 kHz 9 kHz to 10 MHz 10 MHz to 2 GHz 2 GHz to 8 GHz 8 GHz to 20 GHz	0.1 dB 0.065 dB 0.056 dB 0.062 dB 0.087 dB	Network analyzer with calibration kit
ESD Target (Insertion Loss)	5 Hz to 6 GHz	0.1 dB	IEC/EN 61000-4-2,
Adapter (Insertion Loss)	5 Hz to 6 GHz	0.1 dB	ISO 10605
Adapter (Return Loss)	5 Hz to 6 GHz	0.1 dB	Network analyzer with calibration kit
Directional Couplers	9 kHz to 10 MHz 10 MHz to 2 GHz (2 to 20) GHz	0.06 dB 0.04 dB 0.09 dB	Network analyzer with calibration kit
CDNs	9 kHz to 300 MHz	0.27 dB + <i>M</i>	CISPR 16-1-2, IEC/EN 61000-4-6, network analyzer with calibration kit
CDNs (Voltage Division Factor)	9 kHz to 300 MHz	0.27 dB + <i>M</i>	Network analyzer with calibration kit
(50 to 150) Ω Adapters	9 kHz to 1 GHz	0.27 dB + <i>M</i>	IEC/EN 61000-4-6, network analyzer with calibration kit
Current Probes (Transfer Impedance)	5 Hz to 1 GHz	0.24 dB + <i>M</i>	CISPR 16-1-2, IEC/EN 61000-4-6, network analyzer with calibration kit
EM Clamps (Impedance)	9 kHz to 1 GHz	4.2 Ω	IEC/EN 61000-4-6
(Coupling Factor)	9 kHz to 1 GHz	0.27 dB	Network analyzer with calibration kit
(Decoupling Factor)	9 kHz to 1 GHz	0.27 dB	
Decoupling Clamps (Impedance)	9 kHz to 1GHz	4.5 Ω	IEC/EN 61000-4-6
(Decoupling Factor)	9 kHz to 1GHz	0.27 dB	Network analyzer with calibration kit

Parameter/Range	Frequency	CMC ^{2, 4, 5} ($\pm$)	Comments
RF Insertion Loss ³ – (con.)			
LISN, AN, AMN (Voltage Division Factor)	9 kHz to 300 MHz	0.12 dB + <i>M</i>	ANSI C63.4, ISO 7637-2 CISPR 16-1-2, network analyzer with calibration kit
LISN, AN, AMN (Isolation)	9 kHz to 300 MHz	0.17 dB + <i>M</i>	ANSI C63.4, ISO 7637-2 CISPR 16-1-2, network analyzer with calibration kit
CDNE (Voltage Division Factor)	9 kHz to 300 MHz	0.28 dB + <i>M</i>	CISPR 16-1-2, network analyzer with calibration kit
CDNE (Insertion Loss, Isolation)	9 kHz to 300 MHz	0.27 dB + <i>M</i>	CISPR 15 / EN55015, CISPR16-1-2, network analyzer with calibration kit
ISNs (Voltage Division Factor)	9 kHz to 1 GHz	0.61 dB + <i>M</i>	CISPR 22 / EN55022, CISPR 32 / EN55032, CISPR16-1-2, network analyzer with calibration kit
ISNs (Transmission and Cross Talk)	9 kHz to 1 GHz	0.61 dB + <i>M</i>	CISPR 22 / EN 55022, CISPR 32 / EN55032, CISPR 16-1-2, network analyzer with calibration kit
ISNs (LCL)	9 kHz to 1 GHz	0.1 dB + <i>M</i>	CISPR 22 / EN 55022, CISPR 32 / EN55032, CISPR 16-1-2, network analyzer with calibration kit
Amplifiers (Gain)	9 kHz to 20 GHz	0.05 dB	CISPR 22 / EN 55022, CISPR 32 / EN55032, CISPR 16-1-2, network analyzer with calibration kit
Capacitive Voltage Probe	9 kHz to 100 MHz	0.11 dB	Network analyzer with calibration kit CISPR 22 / EN 55022, CISPR 32 / EN55032, CISPR 16-1-2, network analyzer with calibration kit

Parameter/Range	Frequency	CMC ^{2, 4, 5} ($\pm$)	Comments
Impedance ³ –			
LISN, AN, AMN	9 kHz to 1 GHz	0.8 Ω	ANSI C63.4, CISPR 16-1-2 , ISO 7637-2, network analyzer with calibration kit
CDNs	9 kHz to 1 GHz	0.8 Ω	IEC/EN 61000-4-6, network analyzer with calibration kit
ISN	9 kHz to 1 GHz	1.1 Ω	CISPR 22 / EN 55022, CISPR 32 / EN55032, CISPR 16-1-2, network analyzer with calibration kit
Terminators	9 kHz to 10 MHz 10 MHz to 20 GHz	0.13 Ω 0.09 Ω	Network analyzer with calibration kit
Impedance Phase Angle ³ –			
LISN, AN, AMN	9 kHz to 300 MHz	1.1 °	CISPR 16-1-2, network analyzer with calibration kit
ISN	9 kHz to 1 GHz	1.3 °	CISPR 22 / EN 55022, CISPR 32 / EN55032, CISPR 16-1-2, network analyzer with calibration kit
CDNE	9 kHz to 300 MHz	1.3 °	CISPR 15 / EN55015, CISPR16-1-2, network analyzer with calibration kit

Parameter/Range	Frequency	CMC ^{2, 4, 5, 6} ($\pm$)	Comments
Reflection S ₁₁ /S ₂₂ , Magnitude and Phase – (VSWR) Measure ³ (0 to 0.2) lin (0.2 to 0.4) lin (0.4 to 0.6) lin (0.6 to 0.8) lin (0.8 to 1) lin (0 to 0.03) lin (0.03 to 0.04) lin (0.04 to 0.05) lin (0.05 to 0.08) lin (0.08 to 0.09) lin (0.09 to 0.1) lin (0.1 to 0.11) lin (0.11 to 0.12) lin (0.12 to 0.13) lin (0.13 to 0.15) lin (0.14 to 0.15) lin (0.15 to 0.17) lin (0.17 to 0.18) lin (0.18 to 0.19) lin (0.19 to 0.2) lin (0.2 to 0.3) lin (0.3 to 0.4) lin (0.4 to 0.46) lin (0.46 to 1) lin	5 Hz to 9 kHz	(0.0041 to 0.0068) lin (0.0068 to 0.0098) lin (0.0098 to 0.013) lin (0.013 to 0.018) lin (0.018 to 0.022) lin (180 to 8.7) deg (8.7 to 6.7) deg (6.7 to 5.5) deg (5.5 to 3.7) deg (3.7 to 3.4) deg (3.4 to 3.1) deg (3.1 to 2.9) deg (2.9 to 2.7) deg (2.7 to 2.6) deg (2.6 to 2.5) deg (2.5 to 2.3) deg (2.3 to 2.2) deg (2.2 to 2.1) deg (2.1 to 2.0) deg (2.0 to 1.9) deg (1.9 to 1.6) deg (1.6 to 1.4) deg (1.4 to 1.3) deg 1.3 deg	Network analyzer with calibration kit, one port device

Parameter/Range	Frequency	CMC ^{2, 4, 5, 6} ($\pm$)	Comments
Reflection S ₁₁ /S ₂₂ , Magnitude and Phase – (VSWR) Measure ³ (cont)			
(0 to 0.2) lin (0.2 to 0.4) lin (0.4 to 0.6) lin (0.6 to 0.8) lin (0.8 to 1) lin	9 kHz to 10 MHz	(0.0041 to 0.0059) lin (0.0059 to 0.0081) lin (0.0081 to 0.011) lin (0.011 to 0.015) lin (0.015 to 0.019) lin	Network analyzer with calibration kit, one port device
(0 to 0.01) lin (0.01 to 0.1) lin (0.1 to 0.15) lin (0.15 to 0.17) lin (0.17 to 0.18) lin (0.18 to 0.2) lin (0.2 to 0.25) lin (0.25 to 0.3) lin (0.3 to 0.4) lin (0.4 to 0.5) lin (0.5 to 0.6) lin (0.6 to 0.9) lin (0.9 to 1) lin		(180 to 25) deg (25 to 2.8) deg (2.8 to 2.1) deg (2.1 to 1.9) deg (1.9 to 1.8) deg (1.8 to 1.7) deg (1.7 to 1.5) deg (1.5 to 1.3) deg (1.3 to 1.2) deg (1.2 to 1.1) deg (1.1 to 1.0) deg (1.0 to 1.1) deg 1.1 deg	
(0 to 0.2) lin (0.2 to 0.4) lin (0.4 to 0.6) lin (0.6 to 0.8) lin (0.8 to 1) lin	(10 to 500) MHz	(0.0052 to 0.0075) lin (0.0075 to 0.011) lin (0.011 to 0.014) lin (0.014 to 0.018) lin (0.018 to 0.023) lin	
(0 to 0.01) lin (0.01 to 0.1) lin (0.1 to 0.15) lin (0.15 to 0.17) lin (0.17 to 0.18) lin (0.18 to 0.2) lin (0.2 to 0.25) lin (0.25 to 0.3) lin (0.3 to 0.4) lin (0.4 to 0.5) lin (0.5 to 0.6) lin (0.6 to 0.9) lin (0.9 to 1) lin		(180 to 32) deg (32 to 3.6) deg (3.6 to 2.6) deg (2.6 to 2.4) deg (2.4 to 2.3) deg (2.3 to 2.2) deg (2.2 to 1.9) deg (1.9 to 1.7) deg (1.7 to 1.5) deg (1.5 to 1.4) deg (1.4 to 1.3) deg 1.3 deg 1.3 deg	

Parameter/Range	Frequency	CMC ^{2, 4, 5, 6} ($\pm$)	Comments
Reflection S_{11}/S_{22} , Magnitude and Phase – (VSWR) Measure ³ (cont)			
(0 to 0.2) lin (0.2 to 0.4) lin (0.4 to 0.6) lin (0.6 to 0.8) lin (0.8 to 1) lin	500 MHz to 2 GHz	(0.0017 to 0.0033) lin (0.0033 to 0.0050) lin (0.0050 to 0.0070) lin (0.0070 to 0.0091) lin (0.0091 to 0.011) lin	Network analyzer with calibration kit, one port device
(0 to 0.01) lin (0.01 to 0.1) lin (0.1 to 0.15) lin (0.15 to 0.17) lin (0.17 to 0.18) lin (0.18 to 0.2) lin (0.2 to 0.25) lin (0.25 to 0.3) lin (0.3 to 0.4) lin (0.4 to 0.5) lin (0.5 to 0.6) lin (0.6 to 0.9) lin (0.9 to 1) lin		(180 to 11) deg (11 to 1.5) deg (1.5 to 1.2) deg (1.2 to 1.1) deg 1.1 deg (1.1 to 1.0) deg (1.0 to 0.9) deg 0.9 deg (0.9 to 0.8) deg 0.8 deg (0.8 to 0.7) deg 0.7 deg 0.7 deg	
(0 to 0.2) lin (0.2 to 0.4) lin (0.4 to 0.6) lin (0.6 to 0.8) lin (0.8 to 1) lin	(2 to 8) GHz	(0.0023 to 0.0043) lin (0.0043 to 0.0066) lin (0.0066 to 0.0091) lin (0.0091 to 0.012) lin (0.012 to 0.015) lin	
(0 to 0.01) lin (0.01 to 0.1) lin (0.1 to 0.15) lin (0.15 to 0.17) lin (0.17 to 0.18) lin (0.18 to 0.2) lin (0.2 to 0.25) lin (0.25 to 0.3) lin (0.3 to 0.4) lin (0.4 to 0.5) lin (0.5 to 0.6) lin (0.6 to 0.9) lin (0.9 to 1) lin		(180 to 14) deg (14 to 2.2) deg (2.2 to 1.7) deg (1.7 to 1.6) deg (1.6 to 1.5) deg (1.5 to 1.4) deg (1.4 to 1.3) deg (1.3 to 1.2) deg (1.2 to 1.1) deg (1.1 to 1.0) deg (1.0 to 0.9) deg 0.9 deg 0.9 deg	

Parameter/Range	Frequency	CMC ^{2, 4, 5, 6} ($\pm$)	Comments
Reflection S_{11}/S_{22} , Magnitude and Phase – (VSWR) Measure ³ (cont)	(8 to 20) GHz	(0.0046 to 0.0070) lin (0.0070 to 0.0097) lin (0.0097 to 0.013) lin (0.013 to 0.017) lin (0.017 to 0.02) lin (180 to 29) deg (29 to 4.0) deg (4.0 to 3.1) deg (3.1 to 2.9) deg (2.9 to 2.8) deg (2.8 to 2.6) deg (2.6 to 2.2) deg (2.2 to 1.9) deg (1.9 to 1.8) deg (1.1 to 1.0) deg (1.8 to 1.6) deg (1.6 to 1.3) deg (1.3 to 1.1) deg	Network analyzer with calibration kit, one port device

Parameter/Frequency	Range	CMC ^{2, 6} ($\pm$)	Comments
Transmission S ₁₂ /S ₂₁ , Magnitude and Phase – Measure ³			
5 Hz to 9 kHz	(10 to 0) dB	(0.12 to 0.083) dB (0.77 to 0.55) deg	Network analyzer with calibration kit
	(0 to -10) dB	(0.083 to 0.099) dB (0.55 to 0.65) deg	
	(-10 to -20) dB	(0.099 to 0.11) dB (0.65 to 0.75) deg	Non-reflecting device
	(-20 to -30) dB	(0.11 to 0.13) dB (0.75 to 0.85) deg	
	(-30 to -40) dB	(0.13 to 0.15) dB (0.85 to 0.98) deg	
	(-40 to -50) dB	(0.15 to 0.18) dB (0.98 to 1.2) deg	
	(-50 to -60) dB	(0.18 to 0.24) dB (1.2 to 1.6) deg	
	(-60 to -70) dB	(0.24 to 0.40) dB (1.6 to 3.2) deg	
	(-70 to -80) dB	(0.40 to 1.2) dB (3.2 to 8.3) deg	
	(-80 to -90) dB	(1.2 to 3.0) dB (8.3 to 25) deg	

Parameter/Frequency	Range	CMC ^{2, 6} ($\pm$)	Comments
Transmission S_{12}/S_{21} , Magnitude and Phase – Measure ³ (cont)			
9 kHz to 10 MHz	(10 to 0) dB	(0.064 to 0.047) dB (0.45 to 0.31) deg	Network analyzer with calibration kit
	(0 to -10) dB	(0.047 to 0.057) dB (0.31 to 0.37) deg	Network analyzer with calibration kit
	(-10 to -20) dB	(0.057 to 0.067) dB (0.37 to 0.44) deg	Non-reflecting device
	(-20 to -30) dB	(0.067 to 0.089) dB (0.44 to 0.59) deg	
	(-30 to -40) dB	(0.089 to 0.12) dB (0.59 to 0.79) deg	
	(-40 to -50) dB	(0.12 to 0.19) dB (0.79 to 1.3) deg	
	(-50 to -60) dB	(0.19 to 0.43) dB (1.3 to 2.9) deg	
	(-60 to -70) dB	(0.43 to 1.2) dB (2.9 to 8.4) deg	
	(-70 to -80) dB	(1.2 to 3.3) dB (8.4 to 27) deg	
	(-80 to -90) dB	(3.3 to 7.7) dB (27 to 180) deg	

Parameter/Frequency	Range	CMC ^{2, 6} ($\pm$)	Comments
Transmission S_{12}/S_{21} Magnitude and Phase – Measure ³ (cont) (10 to 500) MHz	(10 to 0) dB	(0.046 to 0.037) dB (0.31 to 0.25) deg	Network analyzer with calibration kit
	(0 to -10) dB	(0.037 to 0.046) dB (0.25 to 0.30) deg	Network analyzer with calibration kit
	(-10 to -20) dB	(0.046 to 0.059) dB (0.30 to 0.38) deg	Non-reflecting device
	(-20 to -30) dB	(0.059 to 0.074) dB (0.38 to 0.49) deg	
	(-30 to -40) dB	(0.074 to 0.14) dB (0.49 to 0.88) deg	
	(-40 to -50) dB	(0.14 to 0.25) dB (0.88 to 1.67) deg	
	(-50 to -60) dB	(0.25 to 0.52) dB (1.7 to 3.6) deg	
	(-60 to -70) dB	(0.52 to 1.3) dB (3.6 to 9.1) deg	
	(-70 to -80) dB	(1.3 to 3.3) dB (9.1 to 27) deg	
	(-80 to -90) dB	(3.3 to 7.5) dB (27 to 180) deg	

Parameter/Frequency	Range	CMC ^{2, 6} ($\pm$)	Comments
Transmission S_{12}/S_{21} Magnitude and Phase – Measure ³ (cont)			
500 MHz to 2 GHz	(10 to 0) dB	(0.13 to 0.030) dB (1.7 to 0.20) deg	Network analyzer with calibration kit
	(0 to -10) dB	(0.030 to 0.046) dB (0.20 to 0.38) deg	Network analyzer with calibration kit
	(-10 to -20) dB	(0.046 to 0.054) dB (0.38 to 0.46) deg	Non-reflecting device
	(-20 to -30) dB	(0.054 to 0.064) dB (0.46 to 0.52) deg	
	(-30 to -40) dB	(0.064 to 0.075) dB (0.52 to 0.60) deg	
	(-40 to -50) dB	(0.075 to 0.091) dB (0.60 to 0.71) deg	
	(-50 to -60) dB	(0.091 to 0.15) dB (0.71 to 1.07) deg	
	(-60 to -70) dB	(0.15 to 0.26) dB (1.1 to 1.9) deg	
	(-70 to -80) dB	(0.26 to 0.52) dB (1.9 to 3.7) deg	
	(-80 to -90) dB	(0.52 to 1.3) dB (3.7 to 8.9) deg	

Parameter/Frequency	Range	CMC ^{2, 6} ($\pm$)	Comments
Transmission S_{12}/S_{21} Magnitude and Phase – Measure ³ (cont)			
(2 to 8) GHz	(10 to 0) dB	(0.16 to 0.044) dB (3.6 to 0.29) deg	Network analyzer with calibration kit
	(0 to -10) dB	(0.044 to 0.059) dB (0.29 to 0.62) deg	Network analyzer with calibration kit
	(-10 to -20) dB	(0.059 to 0.069) dB (0.62 to 0.71) deg	Non-reflecting device
	(-20 to -30) dB	(0.069 to 0.078) dB (0.71 to 0.78) deg	
	(-30 to -40) dB	(0.078 to 0.089) dB (0.78 to 0.85) deg	
	(-40 to -50) dB	(0.089 to 0.11) dB (0.85 to 0.71) deg	
	(-50 to -60) dB	(0.11 to 0.15) dB (0.71 to 0.94) deg	
	(-60 to -70) dB	(0.15 to 0.23) dB (0.94 to 1.81) deg	
	(-70 to -80) dB	(0.23 to 0.43) dB (1.9 to 3.2) deg	
	(-80 to -90) dB	(0.43 to 0.97) dB (3.2 to 7.1) deg	

Parameter/Frequency	Range	CMC ^{2, 6} ($\pm$)	Comments
Transmission S_{12}/S_{21} Magnitude and Phase – Measure ³ (cont)			
(8 to 20) GHz	(10 to 0) dB	(0.24 to 0.065) dB (8.7 to 0.43) deg	Network analyzer with calibration kit
	(0 to -10) dB	(0.065 to 0.083) dB (0.43 to 1.2) deg	Non-reflecting device
	(-10 to -20) dB	(0.083 to 0.093) dB (1.2 to 1.3) deg	
	(-20 to -30) dB	(0.093 to 0.11) dB (1.3 to 1.4) deg	
	(-30 to -40) dB	(0.1 to 0.11) dB (1.4 to 1.4) deg	
	(-40 to -50) dB	(0.11 to 0.13) dB (1.4 to 1.5) deg	
	(-50 to -60) dB	(0.13 to 0.17) dB (1.5 to 1.8) deg	
	(-60 to -70) dB	(0.17 to 0.26) dB (1.8 to 2.5) deg	
	(-70 to -80) dB	(0.26 to 0.48) dB (2.5 to 3.9) deg	
	(-80 to -90) dB	(0.48 to 1.1) dB (3.9 to 8.0) deg	

III. Time & Frequency

Parameter/Equipment	Range	CMC ^{2, 5} (±)	Comments
Time Interval and Frequency ³ –	0.5 ns to 60 s	3.5 %	Oscilloscope
ESD Simulator ³ Contact Discharge, Rise Time, RC Time Constant	0.5 ns to 60 s	3.5 %	IEC/EN 61000-4-2, IEC/EN 61000-4-2 (1995), ISO 10605, ISO 10605 (2001), oscilloscope using IEC ESD target
EFT/Burst Generator ³ (50, 1 000) Ω Load			
Rise Time	(2 to 7) ns	3.5 %	IEC/EN 61000-4-4, oscilloscope
Impulse Duration	(30 to 170) ns	3.5 %	IEC/EN 61000-4-4, oscilloscope
Burst Duration	(0.5 to 20) ms	3.5 %	IEC/EN 61000-4-4, oscilloscope
Burst Period	(200 to 400) ms	3.5 %	IEC/EN 61000-4-4, oscilloscope
Repetition Frequency	1 kHz to 1 MHz	3.5 %	IEC/EN 61000-4-4, oscilloscope
Surge Generator ³ Open and Short Circuit			
Front Time, Rise Time	(0.3 to 13) μs	3.5 %	IEC/EN 61000-4-5, IEC/EN 61000-4-5 (2005), IEC/EN 610004-12 oscilloscope
Time to Half Value, Duration	(10 to 900) μs	3.5 %	IEC/EN 61000-4-5, IEC/EN 61000-4-5 (2005), oscilloscope
Open Circuit Phase Shifting	(0 to 20) ms	3.5%	IEC/EN 61000-4-5, IEC/EN 61000-4-5 (2005), IEC/EN 610004-12 oscilloscope

Parameter/Equipment	Range	CMC ^{2, 5} (±)	Comments
Time Interval and Frequency ³ (cont) –			
Oscillation Frequency	(100 to 1000) kHz	3.5 %	IEC/EN 61000-4-12, oscilloscope
Voltage Transient Emission ³			
Switching Time	(200 to 400) ns	3.5 %	ISO 7637-2, ISO 7637-2 (2004), oscilloscope
Transient Immunity ³ Surge Pulse and Load Dump Pulse			
Rise Time	0.4 µs to 15 ms	3.5 %	ISO 7637-2, ISO 7637-2 (2004), ISO 16750-2, oscilloscope
Duration	0.1 µs to 700 ms	3.6 %	ISO 7637-2, ISO 7637-2 (2004), ISO 16750-2, oscilloscope
Transient Immunity ³ Burst Pulse			
Rise Time	(3 to 7) ns	3.5 %	ISO 7637-2, ISO 7637-2 (2004), oscilloscope
Duration	(30 to 200) ns	3.5 %	ISO 7637-2, ISO 7637-2 (2004), oscilloscope

Parameter/Equipment	Range	CMC ^{2, 5} (±)	Comments
Time Interval and Frequency ³ (cont) – PQT ³			
Phase Angle	(0 to 359)°	3.5 %	IEC/EN 61000-4-11, oscilloscope
Pulse Rise/Fall Time	(1 to 5) µs	3.5 %	IEC/EN 61000-4-11, oscilloscope
Voltage Dropout Time	20 µs to 200 ms	3.5 %	Oscilloscope
Repetition Time	1 ms to 5 s	3.5 %	Oscilloscope
Frequency ³ – Measure	(40 to 300) kHz	0.017 %	DMM
	40 Hz to 1 MHz	3.5 %	oscilloscope, HV differential probe
	10 Hz to 3 GHz	0.9 Hz	Frequency counter
	9 kHz to 26.5 GHz	0.3 %	Spectrum analyzer

¹ This laboratory offers commercial Calibration service and field Calibration service.

² Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine Calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific Calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific Calibration.

³ Field Calibration service is available for this Calibration and this laboratory meets A2LA R104 – *General Requirements: Accreditation of Field Testing and Field Calibration Laboratories* for these Calibrations. Please note the actual measurement uncertainties achievable on a customer's site can normally be expected to be larger than the CMC found on the A2LA Scope. Allowance must be made for aspects such as the environment at the place of Calibration and for other possible adverse effects such as those caused by transportation of the Calibration equipment. The usual allowance for the actual uncertainty introduced by the item being calibrated, (e.g. resolution) must also be considered and this, on its own, could result in the actual measurement uncertainty achievable on a customer's site being larger than the CMC.

⁴ In the statement of CMC, M is the Mismatch error. Uncertainty does not include mismatch error due to connections of the device to other devices in actual use. Mismatch uncertainties, due to the reflection coefficient of the device to be calibrated, are to be included in the overall measurement uncertainty; the approach of determining expanded uncertainties at approximately the 95% level of confidence, (using a coverage factor of $k = 2$) is to be applied for this calculation as well.

⁵ In the statement of CMC, the value is defined as the percentage of reading unless otherwise noted.

⁶ CMC for intermediate values of measurand can be found by interpolation.



Accredited Laboratory

A2LA has accredited

AMETEK CO., LTD. NAGOYA OFFICE

Nagoya-shi, Aichi-ken, JAPAN

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets the requirements of ANSI/NCSLI Z540-1-1994 and any additional program requirements in the field of calibration. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009).



Presented this 8th day of April 2015.

A handwritten signature in blue ink, appearing to read 'J. C. Bunt'.

Senior Director of Quality and Communication
For the Accreditation Council
Certificate Number 2127.03
Valid to April 30, 2017
Revised on February 3, 2016

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.